Vote&Check: a verifiable postal voting protocol

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06/05/2025

Accepted paper at PETS '25



Motivation

Why being interested in postal voting?

Widespread in practice

Postal voting is widely used:

- Professional elections
- Trade union elections
- Associations
- University boards of directors
- Political primaries



Also for high-stakes elections



2022 Conservative Party leadership election

Current situation

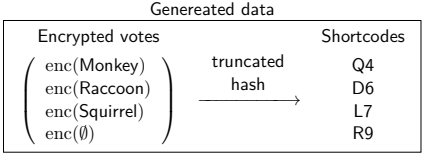
There is less interest in postal voting than in electronic voting.

- STROBE [Benaloh, E-Vote-ID'21]
- RemoteVote & SAFE Vote [Crimmins, Rhea, Halderman, FC'22]
- Louvain [Devillez, Pereira, Peters, 2024]

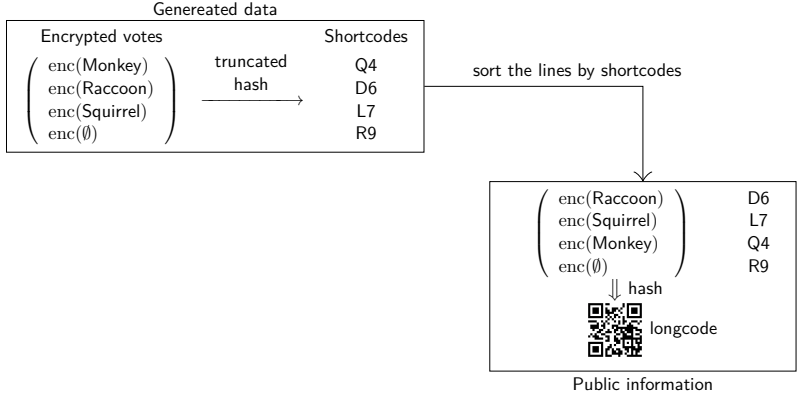
Why?

- Electronic channels are more familiar.
- E-voting was suppose to replace postal voting (in practice: complement it).
- Sounds oldschool.

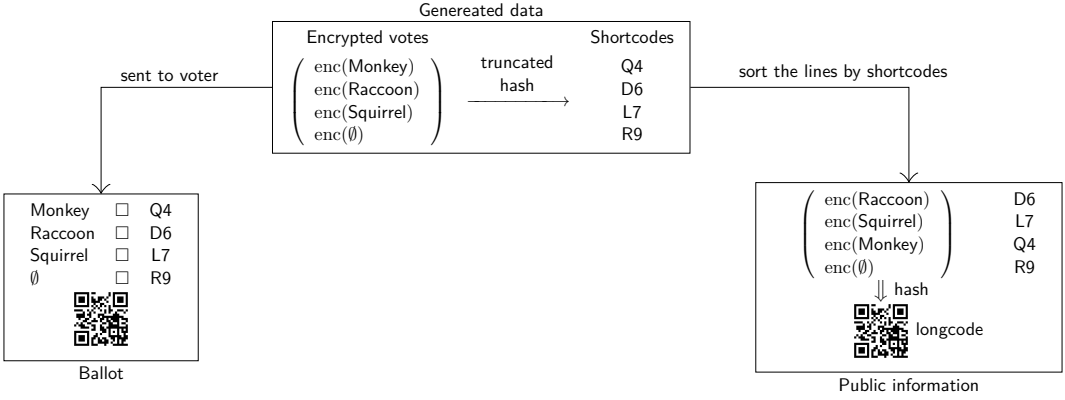
STROBE in a nutshell



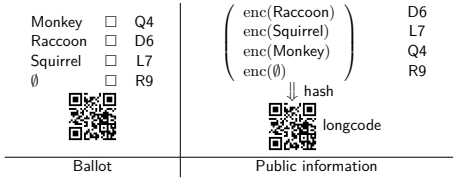
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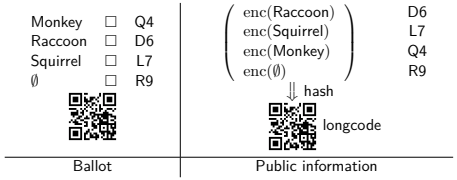
Printer

Voter

Public board

Polling station

STROBE in a nutshell



Printer

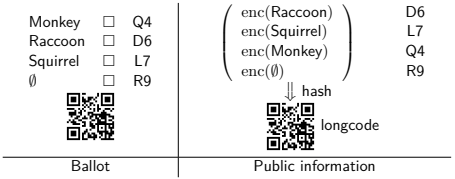
Public data + ZKP

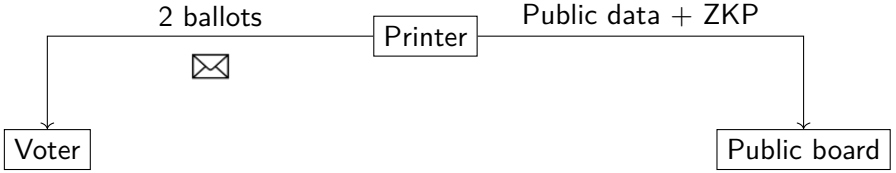
Voter

Public board

Polling station

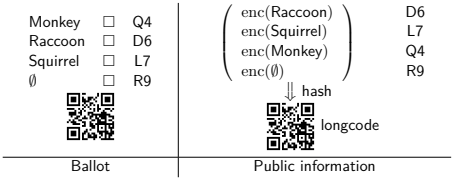
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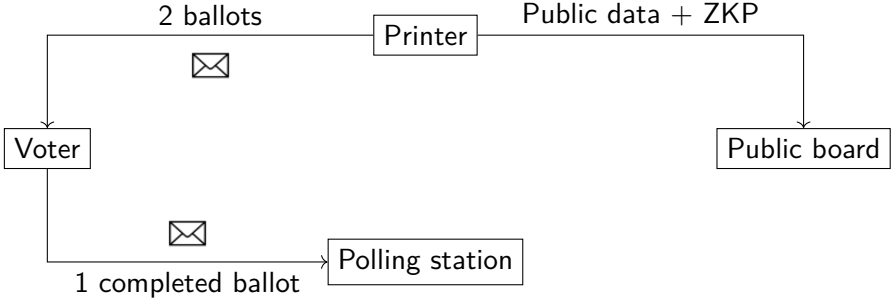




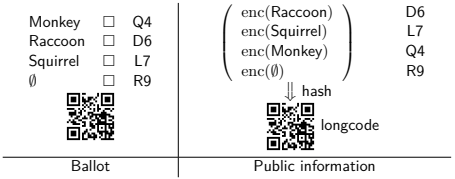
Polling station

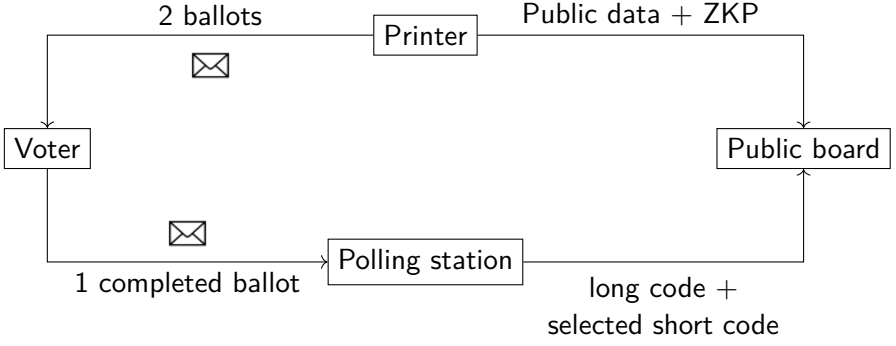
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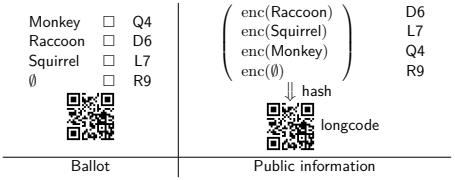


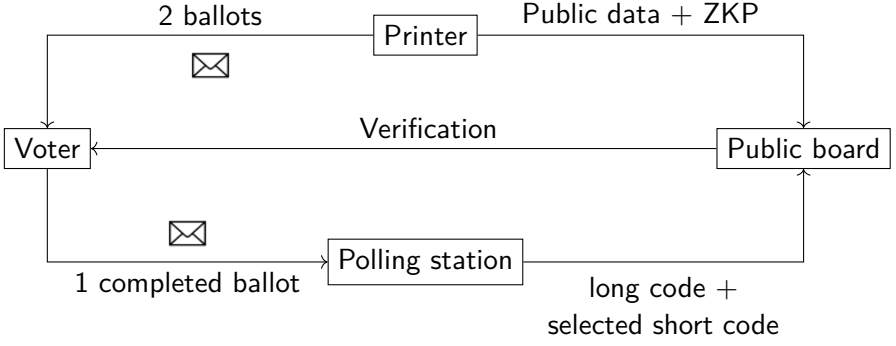
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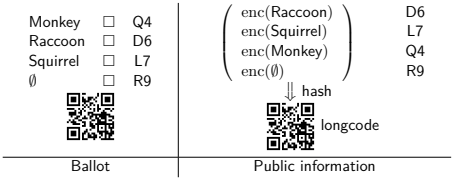


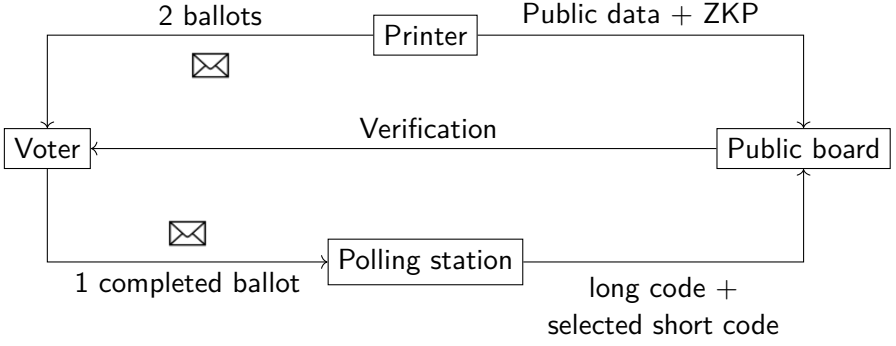
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STROBE in a nutshell



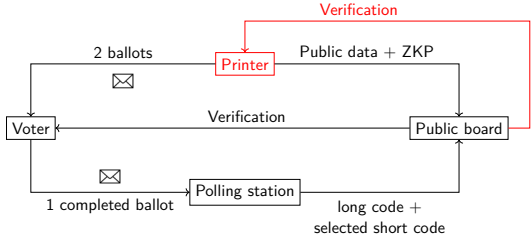


- Chosen encrypted votes are added and homomorphically decrypted
- Uncast ballots are spoilt and audited

Attacks from a dishonest printer

Full privacy breach

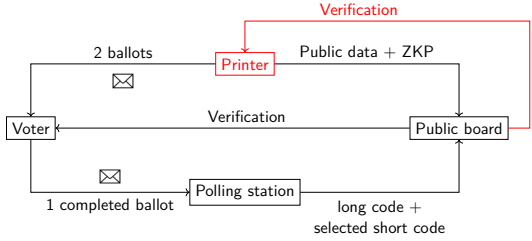
1. Printer emits material as usual, and keeps the ballot-voter link
2. Voters vote as usual
3. Printer performs verification instead of the voters



Attacks from a dishonest printer

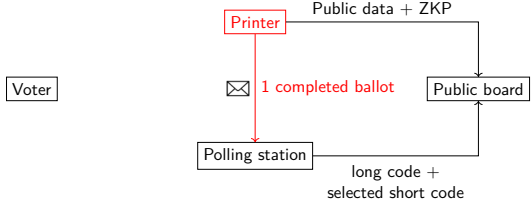
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Weak eligibility

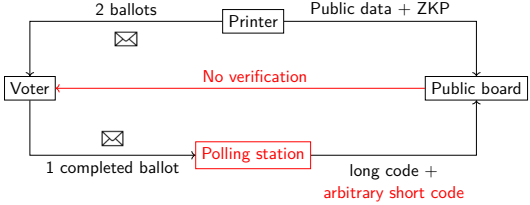
1. Printer emits material as usual and keeps copies of the ballots
2. Printer uses the copy to vote instead of the abstainers



Attacks from a dishonest polling station

Alter non-verifying voters

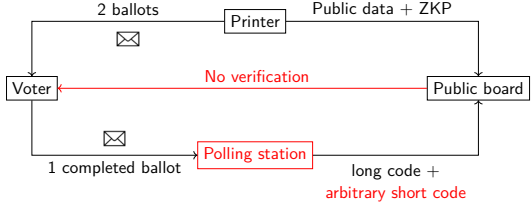
1. Polling station can record votes dishonestly
2. No one verifies for a voter who does not



Attacks from a dishonest polling station

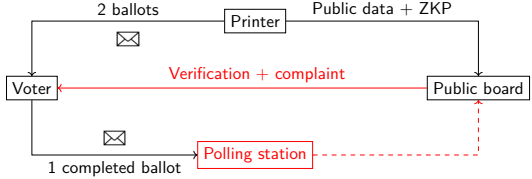
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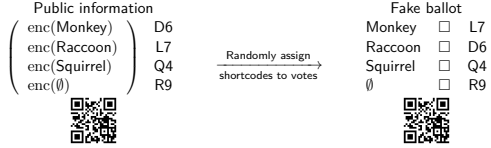
Privacy breach by complaints

1. Polling station drops some ballots voting Monkey
2. Voters complaining have voted Monkey



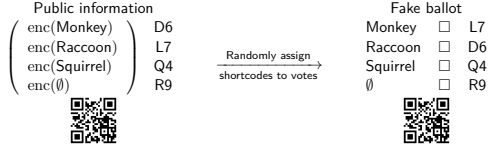
Attacks from a third party

Forging fake ballots



Attacks from a third party

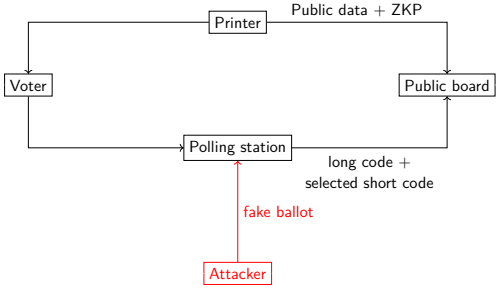
Forging fake ballots



Vote flooding

1. Attacker creates fake ballots and casts them
2. Random votes are recorded

Useful to reach a threshold, or to increase turnout.



Generic attacks

		STROBE [1]	RemoteVote [2]	SAFE Vote [2]	Louvain [3]	**Vote&Check**
	# of authorities	3	3	3	5	3
privacy attacks	full privacy breach	😈	😈	😈	🛡	🛡
	targeted privacy breach	😈	😈	😈	🛡	🛡
	privacy breach by complaints	😈	😈	😈	😈	😈
verifiability attacks	vote flooding	😈	😈	🛡	🛡	🛡
	clash attack	🛡	😈	😈	🛡	🛡
	alter non verifying voters	😈	😈	😈	😈	😈
	weak eligibility	😈	😈	😈	😈	😈
	universal verifiability	🛡	🛡	🛡	proxy	🛡

[1] Benaloh. 2021. STROBE-Voting: Send Two, Receive One Ballot Encoding

[2] Crimmins, Rhea, Halderman. 2022. RemoteVote and SAFE Vote: Towards Usable End-to-End Verification for Vote-by-Mail

[3] Devillez, Pereira, Peters. 2024. Verifiable and Private Vote-by-Mail

New proposal: Vote&Check

Goal: design a verifiable postal voting protocol with security proofs

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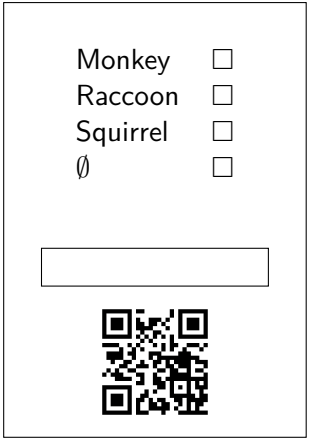
Goal: design a verifiable postal voting protocol with security proofs

Constraints:

- Possible to vote "as usual"
- At least as good as state of the art

POV: you are the voter

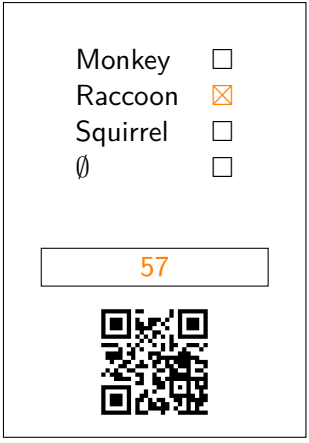
Step 1: receiving by post





POV: you are the voter

Step 2: voting

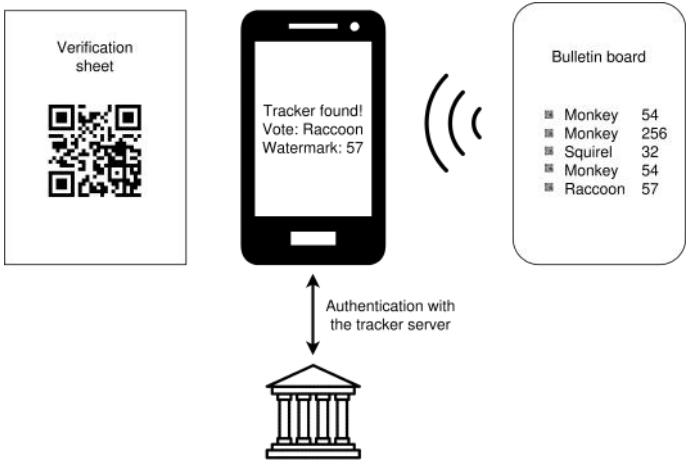


Voter selects a vote

And randomly chooses a short number

POV: you are the voter

Step 3: checking (optional)



Design

Tracker Server

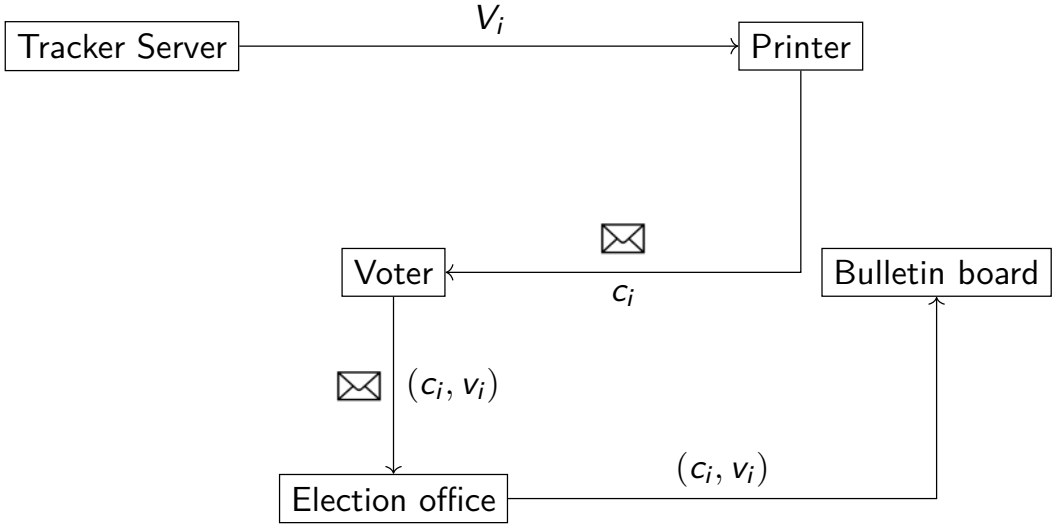
Printer

Voter

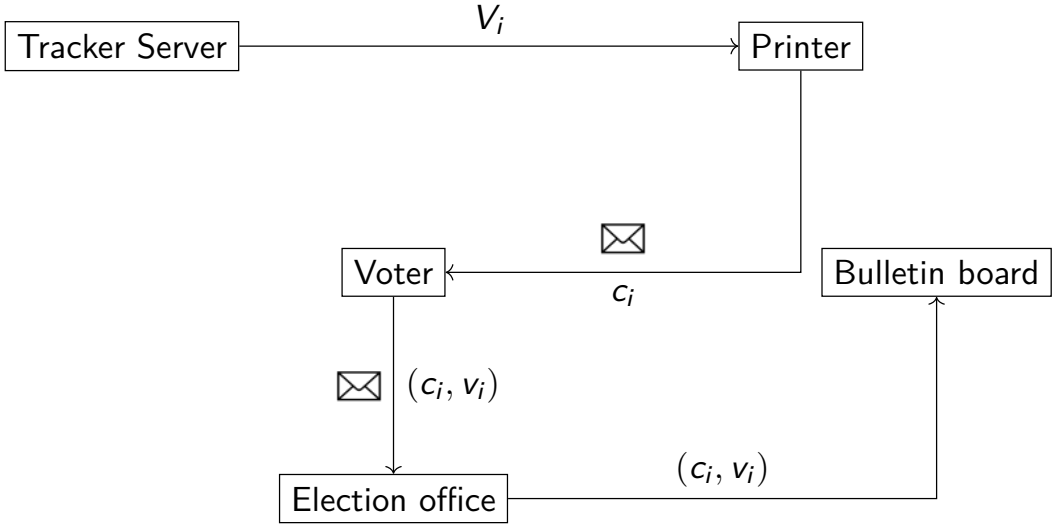
Bulletin board

Election office

Design

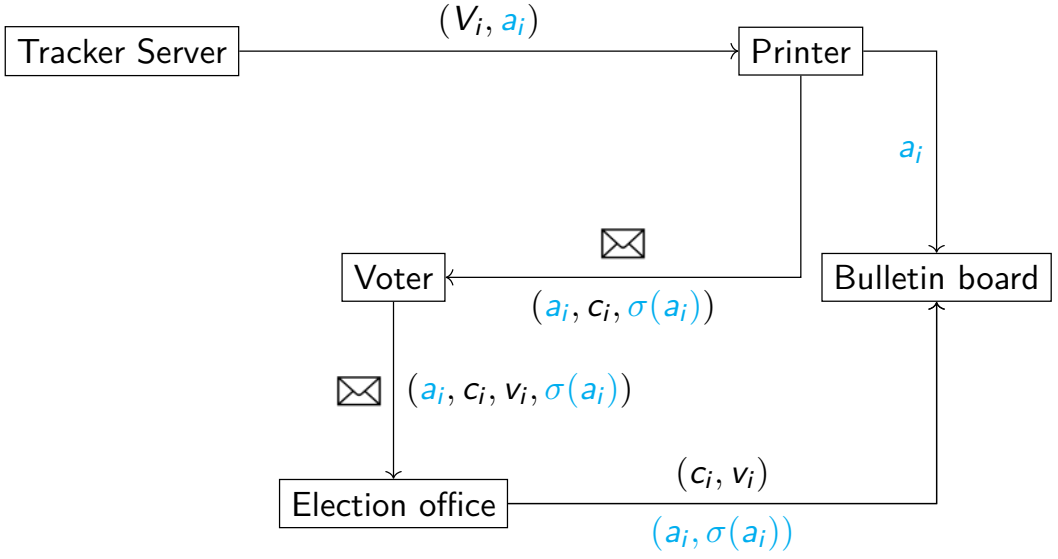

Tracker Server
V_i
Printer
Voter
c_i
Bulletin board
(c_i, v_i)
Election office
(c_i, v_i)

Design

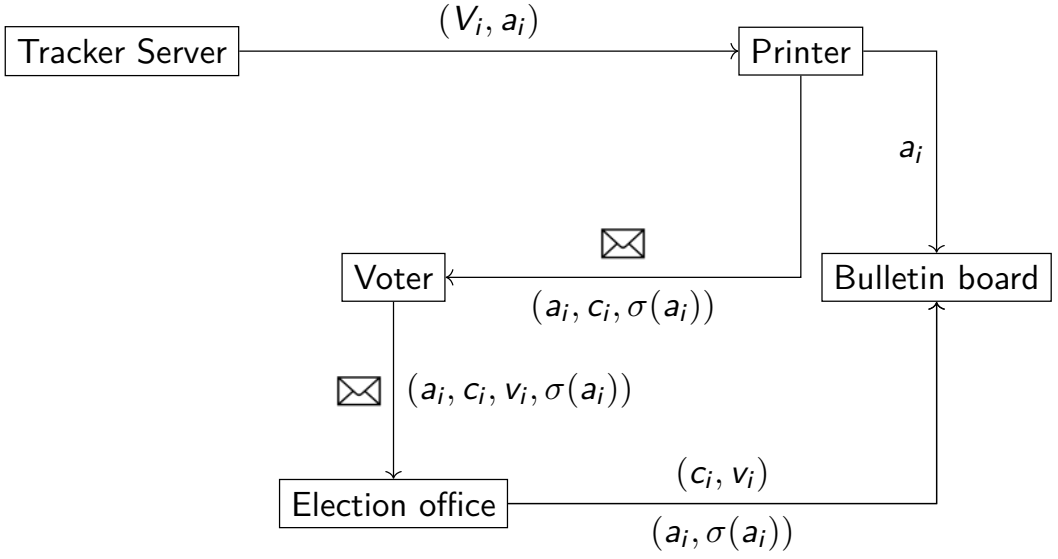


⚠ Issue: anyone can vote any number of times.

Design

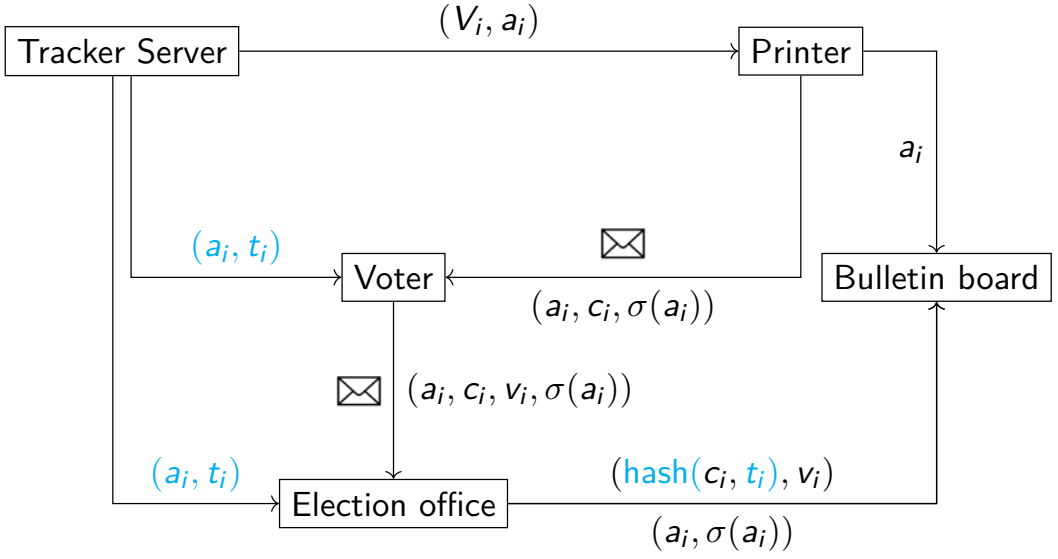

Tracker Server
(V_i, a_i)
Printer
a_i
Voter
$(a_i, c_i, \sigma(a_i))$
Bulletin board
$(a_i, c_i, v_i, \sigma(a_i))$
Election office
(c_i, v_i)
$(a_i, \sigma(a_i))$

Design

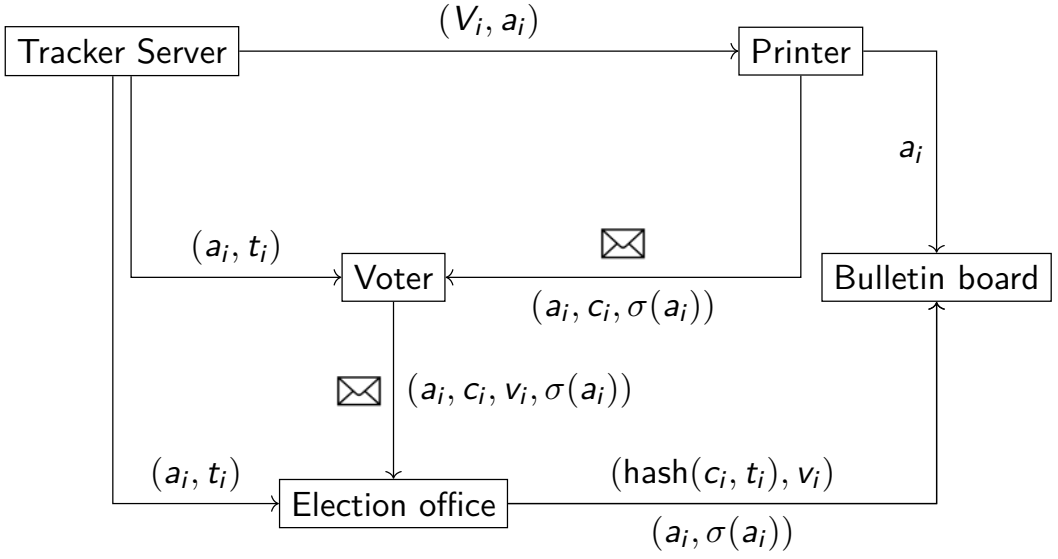


⚠ Issue: the printer learns all the votes.

Design

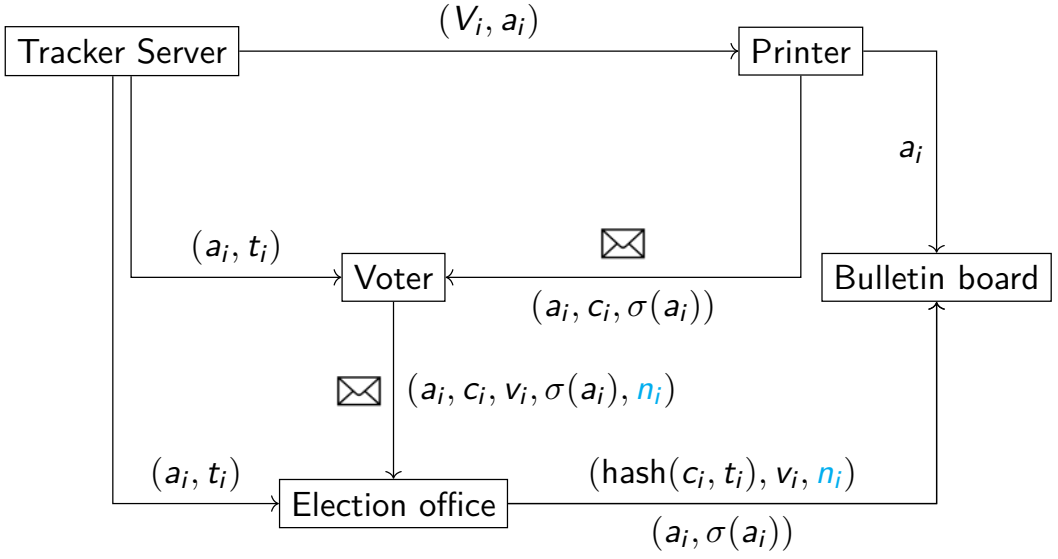

Tracker Server
(V_i, a_i)
Printer
a_i
(a_i, t_i)
Voter
$(a_i, c_i, \sigma(a_i))$
Bulletin board
$(a_i, c_i, v_i, \sigma(a_i))$
(a_i, t_i)
Election office
$(\text{hash}(c_i, t_i), v_i)$
$(a_i, \sigma(a_i))$

Design



Issue: clash attack

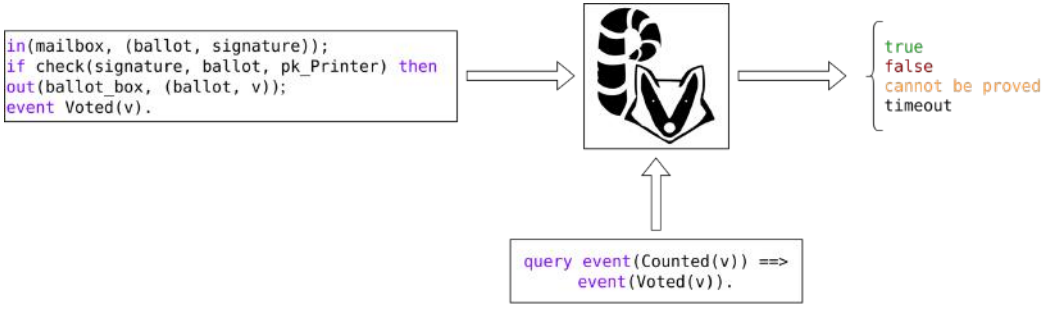
Design


Tracker Server
(V_i, a_i)
Printer
a_i
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Security analysis

Protocole pas prouvé, protocole pété

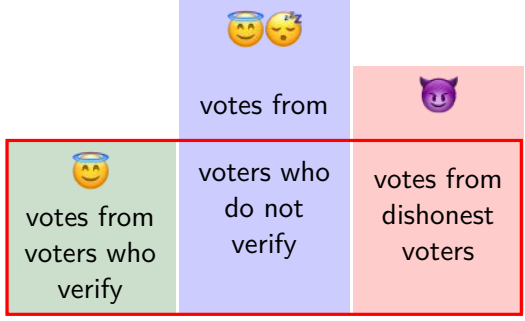
The Proverif tool



End-to-end verifiability

In a few words: the attacker cannot change the outcome of the election.

Result of the election =



End-to-end verifiability

Individual verifiability

```
query Finish && Inj-Verified(vote) ==> Inj-Counted(vote)
```

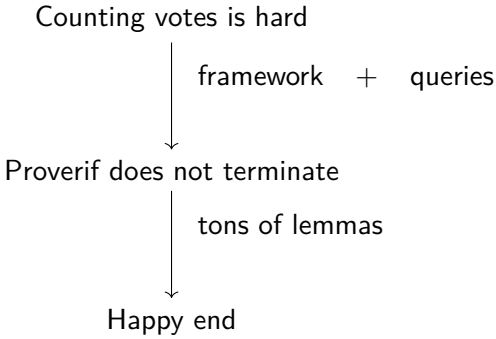
Strong eligibility

```
query Finish && Inj-Counted(vote) ==>
   Inj-Honest(Voter) && Verified(Voter, vote)
   || Inj-Lazy(Voter) && Voted(Voter, vote)
   || Inj-Corrupt(Voter)
```

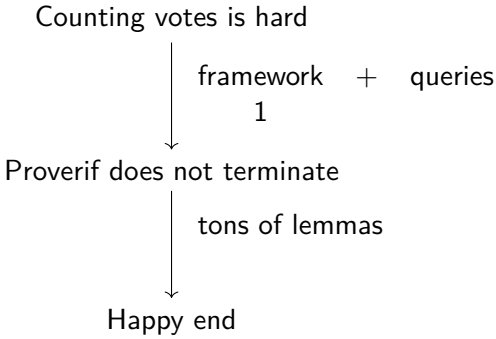
Theorem [Cheval, Cortier, Debant, CSF'23]

Individual verifiability + Strong eligibility $\Longrightarrow$ E2E-Verifiability

Using the verifiability framework [ibid.]

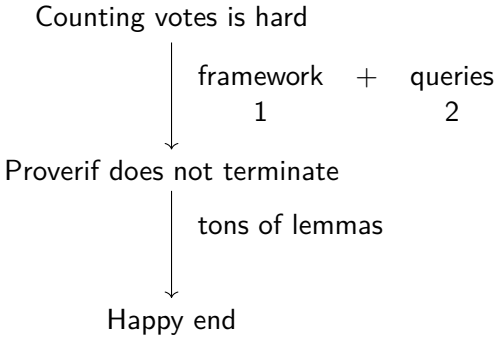


Using the verifiability framework [ibid.]



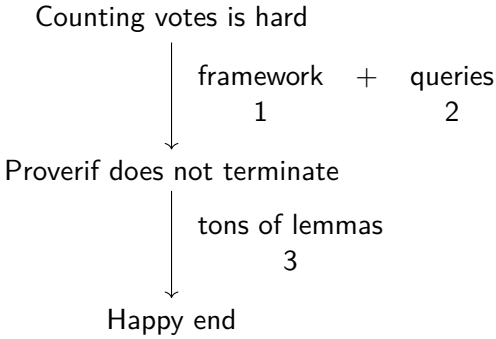
1. Fit our protocol in the framework

Using the verifiability framework [ibid.]



1. Fit our protocol in the framework
2. Define a new property

Using the verifiability framework [ibid.]



1. Fit our protocol in the framework
2. Define a new property
3. Introduce new lemmas + a pen-and-paper axiom

Voter ident

- Link a ballot to a voter
- Ballots do not contains a natural identifier in Vote&Check
- Which nonce brings security depends of the corruption scenario

4 lines of code for Belenios, 50 for Vote&Check

Weak verifiability

Issue: E2E-Verifiability is too strong for postal voting.

Strong eligibility (from the framework)

```
query Finish && Inj-Counted(vote) ==>
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```

Weak eligibility (our property)

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      || Inj-Corrupt(Voter)
```

Weak verifiability

Issue: E2E-Verifiability is too strong for postal voting.

Strong eligibility (from the framework)

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Security claims

Tracker Server	Printer	Election Office	Privacy	E2E Verifiability	Weak verifiability	Individual verifiability
H	H	H	✓	✓	✓	✓
C	H	H	✓	✓	✓	✓
H	C	H	✓	✗	✗	✓
H	H	C	✓	✗	✓	✓
H	C	C	✗	✗	✗	✓
C	H	C	✗	✗	✓	✓
C	C	H	✗	✗	✗	✓
C	C	C	✗	✗	✗	✓

- Privacy holds if at most one authority is dishonest.
- Individual verifiability is unconditional.
- Dishonest election office can modify votes from lazy voters.
- Dishonest printer can cast ballots for abstainers.

Conclusion

Generic attacks on existing protocols

- Breaking both privacy and verifiability
- Set realistic objectives

New protocol: Vote&Check

- Only simple cryptography
- Reduced number of authorities

Formal proofs

- New definitions
- Proverif models